

STAT 709: Homework 2

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1. Textbook exercises

Jun Shao's book, Chapter 1: Exercises 35(a)–(b), 40, and 55 (required); Exercise 83 (optional).

2. Integration by parts (optional)

Let $a < b$ and $f, g \in C^1([a, b])$, the space of continuously differentiable functions on $[a, b]$. The usual integration-by-parts formula is

$$\int_a^b f'(x)g(x) dx = f(b)g(b) - f(a)g(a) - \int_a^b f(x)g'(x) dx.$$

We can regard $f'(x) dx$ and $g'(x) dx$ as defining signed measures on $[a, b]$. If the derivatives are nonnegative, these are nonnegative measures. We now generalize the formula to a probability measure with cdf F . Recall that probability measures on $\mathbb{R}$ correspond one-to-one to cdfs. For $g \in C^1([a, b])$, prove that

$$\int_{(a,b]} g(x) dF(x) = F(b)g(b) - F(a)g(a) - \int_a^b g'(x)F(x) dx.$$

Hint: Express $g(x)$ as an integral and apply Fubini's theorem.

3. Moment bounds

Let X be a real-valued random variable.

(1) Suppose $0 < p < q$ and $\mathbb{E}|X|^q < \infty$. Use Hölder's inequality to show that $\mathbb{E}|X|^p < \infty$ and

$$(\mathbb{E}|X|^p)^{1/p} \leq (\mathbb{E}|X|^q)^{1/q}.$$

This says that a higher moment condition is more restrictive, as we will see in the weak law of large numbers.

(2) Suppose $M = \sup_{p \geq 1} (\mathbb{E}|X|^p)^{1/p} < \infty$. Prove that $\mathbb{P}(|X| \leq M) = 1$.

(3) (Optional) Let $G \sim N(0, 1)$. Prove that there is a constant $c > 0$ such that

$$\lim_{n \rightarrow \infty} \frac{(\mathbb{E}|G|^n)^{1/n}}{\sqrt{n}} = c,$$

and determine c .